

LAB 4: Logical, Physical Data Models and SQL Code

Due Date: None

Published: 23rd March 2026

Exercise	Topic	Marks
Exercise 1	Healthcare Clinic System	15
Exercise 2	Corporate Staff System	15
Exercise 3 (Submission)	Retail Loyalty System	20
	TOTAL	50

General Instructions

- A conceptual and logical models were discussed in **Lab 3**, for exercise 1 and 2. The objective was to introduce you Entity Relationship diagrams using www.draw.io).
- This lab aims to reinforce your understanding of logical and physical data models as well the implementation thereof, for both exercise 2 and 3.
- The logical model is about placing foreign keys: in a 1:1 relationship, a foreign key can be placed on any side, 1:M, the foreign key is placed on the many side, M:N relationship, you have create an associative relationship that will have the two primary keys from the M:N relationship and each primary key will also be a foreign key referencing back to the entities with the primary keys.
- Each entity must have a clearly identified **primary key**, marked with **PK** and underlined inside the entity box in Crow's Foot notation.
- Each associative entity must have both identifying **primary keys** (PK), and foreign keys FK.
- **You are required to draw a logical, physical data model and implement your physical data model in SQL Server using SQL Server Management Studio, for both exercise 2 and 3.**

Exercise 1 — Healthcare Clinic System

[15 Marks]

Given the following business rules for a Healthcare Clinic, identify all the entities. Choose **at least 5 attributes** for each entity and include a primary key. Draw the ERD using Crow's Foot Notation.

- a) A Doctor can treat many Patients. **[2 marks]**
- b) A Patient can be treated by many Doctors. **[2 marks]**

- c) A Doctor can be assigned to multiple Wards. [2 marks]
 - d) A Ward can accommodate multiple Doctors on rotation. [2 marks]
 - e) Correctly identified entities with appropriate attributes (min. 5 each) and primary keys. [2 marks]
 - f) Relationships correctly labelled with cardinalities. [5 marks]
-

Exercise 2 — Corporate Staff Management System

[15 Marks]

Given the following business rules for a Corporate HR system, identify all the entities. Choose **at least 4 attributes** for each entity and include a primary key. Draw the ERD using Crow's Foot Notation.

- a) An Employee can report to one or more Managers. [2 marks]
 - b) A Manager can oversee one or more Employees. [2 marks]
 - c) An Employee can mentor multiple Junior Staff members. [2 marks]
 - d) A Junior Staff member can be mentored by several Employees. [2 marks]
 - e) Correctly identified entities with appropriate attributes (min. 4 each) and primary keys. [2 marks]
 - f) Relationships correctly labelled with cardinalities. [5 marks]
-

Exercise 3 — Retail Loyalty Programme System

[20 Marks]

Given the following business rules for a Retail Loyalty Programme, identify all the entities. Choose **at least**

4 attributes for each entity and include a primary key. Complete your ERD in **draw.io**

(<https://app.diagrams.net/>) using Crow's Foot Notation and **submit the diagram as a PDF on eLearning.**

- a) A Customer can enrol in one Loyalty Programme. [2 marks]
 - b) A Loyalty Programme can have many Customers enrolled. [2 marks]
 - c) A Customer can make multiple Purchases. [2 marks]
 - d) Each Purchase is associated with one specific Customer. [2 marks]
 - e) A Customer can maintain multiple Wishlists. [2 marks]
 - f) Each Wishlist belongs to only one Customer. [2 marks]
 - g) Correctly identified entities with appropriate attributes (min. 4 each) and primary keys [4 marks].
 - h) Relationships correctly labelled with cardinalities using Crow's Foot notation. [2 marks]
-